

BO-YU CHEN

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EDUCATION

National Taiwan University (NTU)

Taipei, Taiwan

B.S. in Physics & Computer Science | GPA: 4.09/4.30

Sep 2023 – June 2027 (expected)

- Recipient of *Fu Bell Scholarship* — highest 4-year merit award for top 1% of NTU; renewable for academic excellence.
- First Place, Physics department's undergraduate special talent admission.
- Computer Science double major, selected among 6 university-wide admits per year

University of Illinois at Urbana–Champaign (UIUC)

Urbana, USA

Exchange Student | GPA: 4.00/4.00 (32 credits, all in PHYS)

Sep 2025 – May 2026

- Supported by Chuan-Pu Lee Memorial Scholarship and full tuition waiver from UIUC.
- Selected course: PHYS 514 Modern Atomic Physics (**Grad-level, A+, Rank 1/22 in final presentation**)
- *Dean's List*, Grainger College of Engineering (Fall 2025, Spring 2026)

PUBLICATIONS

*Equal contribution. **Citations Summary:** h-index = 4, Total citations = 215 ([Google Scholar](#))

- [7] *atommovr*: An open-source simulation framework for rearrangement in atomic arrays,

Nikhil K. Harle*, **Bo-Yu Chen***, Bob Bao, Hannes Bernien,

Quantum **10**, 2177, 2026.

This work was highlighted in *Quantum Zeitgeist*.

Community adoption: Publicly forked or/and extended by AMO groups at MIT (Zwierlein), LMU/MPQ (Aidelsburger, Zeiher). External use includes graduate **thesis work** in Aidelsburger group, and active **camera-AWG-AOD integration** in the Zwierlein group.

- [6] Nonparametric Modern Hopfield Models,

Jerry Yao-Chieh Hu*, **Bo-Yu Chen***, Dennis Wu, Feng Ruan, Han Liu,

In *42nd International Conference on Machine Learning (ICML'25)*, 2025. [arXiv:2404.03900](#)

- [5] STanHop: Sparse Tandem Hopfield Model for Memory-Enhanced Time Series Prediction,

Dennis Wu*, Jerry Yao-Chieh Hu*, Weijian Li*, **Bo-Yu Chen**, Han Liu,

In *12th International Conference on Learning Representations (ICLR'24)*, 2024. [arXiv:2312.17346](#)

- [4] Magnetoresistance Properties in Nickel-catalyzed, Air-stable, Uniform, and Transfer-free Graphene,

Bo-Yu Chen, Bo-Wei Chen, Wu-Yih Uen, Chi Chen, Chiashain Chuang, Dung-Sheng Tsai,

Nanotechnology **35**, 205706, 2024.

- [3] On Sparse Modern Hopfield Model,

Jerry Yao-Chieh Hu, Donglin Yang, Dennis Wu, Chenwei Xu, **Bo-Yu Chen**, Han Liu,

In *37th Conference on Neural Information Processing Systems (NeurIPS'23)*, 2023. [arXiv:2309.12673](#)

This work was highlighted in *Northwestern CS department news*.

- [2] Modulations for Quantum Electronic Material Transports by Vacuum Annealing Methods,

Ji-Wei Ci, **Bo-Yu Chen**, Yuan-Chih Hung, Huan-Chien Wang, Dung-Sheng Tsai, Wu-Yih Uen, Yuan-Liang Zhong,

Jhy-Shyang Wang, Chi-Te Liang, Chiashain Chuang,

Spin **13**, 2340023, 2023.

- [1] First-Principles Study on Possible Half-Metallic Ferrimagnetism in Double Perovskites $\text{Pb}_2\text{XX}'\text{O}_6$ ($\text{X} = \text{Ti, Zr, Hf, V, Nb}$ and Ta , $\text{X}' = \text{Tc, Ru, Os}$ and Rh),

Bo-Yu Chen, Po-Han Lee, Yin-Kuo Wang,

Materials **15**, 3311, 2022.

AWARDS & HONORS

- **Fu Bell Scholarship**, NTU, Taiwan 2023 - 2026 (6 consecutive semesters)
- **College of Science Travel Grants and Scholarship**, College of Science, NTU, Taiwan 2024, 2026

- **Dean's List**, Grainger College of Engineering, UIUC, USA 2025 Fall, 2026 Spring
- **School of Computing Global Preview Week (23 selected from 1,000+ applicants)**, KAIST, Korea 2026 (declined)
- **Fermilab Quantum Symposium (official invitation only)**, Fermilab, USA 2025
- **MCQST Summer Bachelor Program**, Munich Center for Quantum Science and Technology, Germany 2025
- **Chuan-Pu Lee Memorial Scholarship**, College of Science, NTU, Taiwan 2025
- **UChicago-Taiwan Student Exchange (UCTS) Fellowship**, Department of Physics, UChicago, USA 2024
- **Dean's List** (Awarded to students ranking in the top 5%), NTU, Taiwan 2024 Spring
- **Taipei City Mayor Award**, Taipei City Government, Taiwan 2023
- **Sakura Science Exchange Program**, invited by Japan Science and Technology Agency, Japan 2023

RESEARCH EXPERIENCE

Department of Physics, Columbia University

New York, USA

Research Intern (Advisor: Prof. Zhenjie Yan)

May 2026 – Feb 2027 (expected)

- Constructing a lithium spectroscopy vapor cell and vacuum system
- Setting up laser distribution system for tapered amplifier and spectroscopy experiment
- Building a National Instruments-based laboratory control system
- Investigating collective decay of Rydberg atoms on QuEra quantum hardware

Department of Physics & James Franck Institute, University of Chicago

Chicago, USA

Research Intern (Advisor: Prof. Cheng Chin; Co-advisor: Prof. Jacob P. Covey)

Sep 2025 – May 2026

- Implemented FPGA for memory controller and trigger logic for controlling DMD (DLP 7000)
- Reverse-engineered Texas Instrument's source code for controlling DMD (DLP 4500)
- Visited Prof. Chen-Lung Hung group's at Purdue to upgrade DMD (DLP650 LNIR) for dynamic patterns control and benchmark performance.
- Designed a firmware-level translation layer that converts *atommovr* trajectories into compressed binary streams.
- Investigated methods for generating uniform intensity profile beam (phase diffuser / microlens array / square-core fiber)

Max-Planck-Institut für Quantenoptik

Garching, Germany

Research Intern (Advisor: Prof. Johannes Zeiher)

Jul 2025 – Aug 2025

- Supported by MCQST Summer Bachelor Program; 16 selected from 312 worldwide applicants.
- Built crossed-AODs experiment for testing application of *atommovr* in real-time atom rearrangement control.

Institut für Quantenoptik und Quanteninformation

Innsbruck, Austria

Research Intern (Advisor: Prof. Hannes Bernien)

Dec 2024 – Feb 2025

- Supported by IQOQI Innsbruck and NTU travel grants.
- Proposed the first algorithm achieving unity success rate in preparing arbitrary targets for dual-species atom arrays. [7]

Pritzker School of Molecular Engineering, University of Chicago

Chicago, USA

Research Intern (Advisor: Prof. Hannes Bernien)

Jun 2024 – Sep 2024

- Supported by UCTS Fellowship; youngest recipient; **featured in UChicago news**.
- Developed *atommovr*, the first open-source package for simulating rearrangement in atomic arrays. [7]

Department of Computer Science, Northwestern University

Evanston, USA

Remote Research Intern (Advisor: Prof. Han Liu)

Jan 2023 – Jan 2024

- Conducted theoretical research on pretrained models through the framework of dense associative memory. [3] [5] [6]

CONFERENCE PRESENTATIONS

- [4] **Coherent Reactions and Dynamics in Cs₂ Molecular Quantum Gas**, Chuixin Kong, Shu Nagata, **Bo-Yu Chen**, Tadej Meznarsic, Cheng Chin, 57th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Providence, RI, June 2026
- [3] **ExpAtommovr: Experimental Setup for Atom Rearrangement Control**, **Bo-Yu Chen**, Adrien Bouscal, Johannes Zeiher, MCQST Symposium, Garching, Germany, August 2025
- [2] **Dual-Species Neutral Atom Array 2.0: Towards a Versatile and Scalable Quantum Processor**, Bob Bao, Nikhil Harle, **Bo-Yu Chen**, Jinyue Jiang, Hannes Bernien, Quantum Science Gordon Research Conference, MA, USA, July 2024

- [1] Temperature-Dependent Magnetoresistance of Transfer-Free Graphene Grown by APCVD,
Bo-Yu Chen, Bo-Wei Chen, Ji-Wei Ci, Wu-Yih Uen, Po-Han Lee, Chi Chen, Chiashain Chuang, Dung-Sheng Tsai,
13th Recent Progress in Graphene and 2D Materials Research Conference, Taipei, Taiwan, November 2022